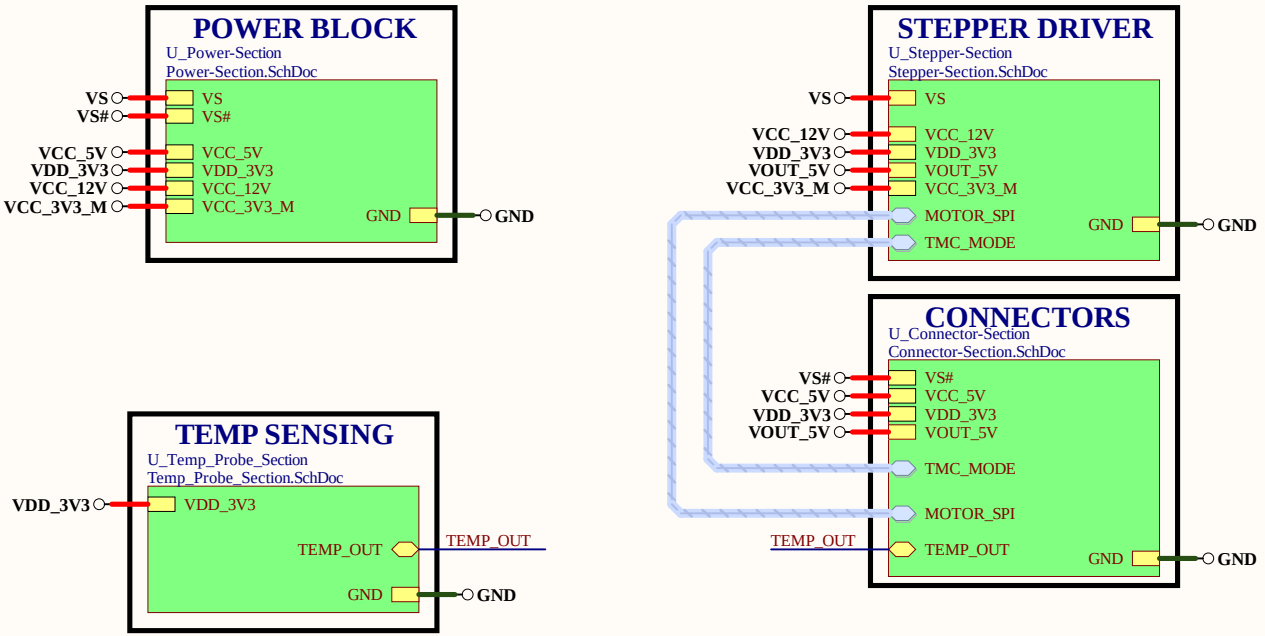
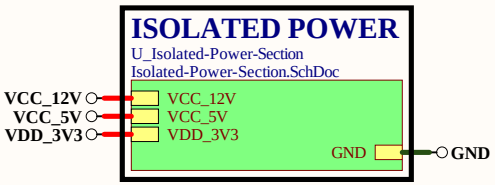
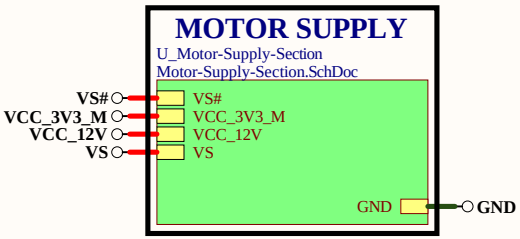
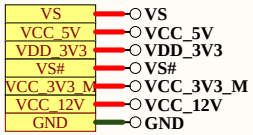
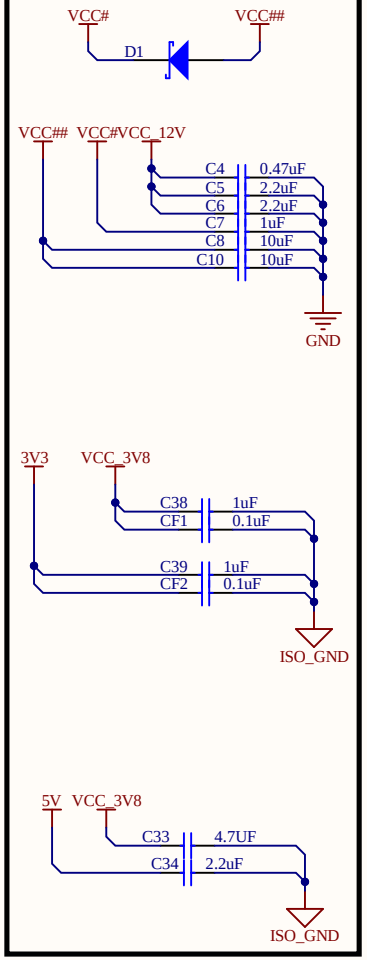

POWER BLOCK
U_Power-Section
Power-Section.SchDoc
VS
VS#
VCC_5V
VDD_3V3
VCC_12V
VCC_3V3_M
GND
GND
STEPPER DRIVER
U_Stepper-Section
Stepper-Section.SchDoc
VS
VCC_12V
VDD_3V3
VOUT_5V
VCC_3V3_M
MOTOR_SPI
TMC_MODE
GND
GND
CONNECTORS
U_Connector-Section
Connector-Section.SchDoc
VS#
VCC_5V
VDD_3V3
VOUT_5V
TMC_MODE
MOTOR_SPI
TEMP_OUT
GND
GND
TEMP SENSING
U_Temp_Probe_Section
Temp_Probe_Section.SchDoc
VDD_3V3
TEMP_OUT
GND
GND


ISOLATED POWER
U_Isolated-Power-Section
Isolated-Power-Section.SchDoc
VCC_12V
VCC_5V
VDD_3V3
GND
VCC_12V
VCC_5V
VDD_3V3
GND
GND

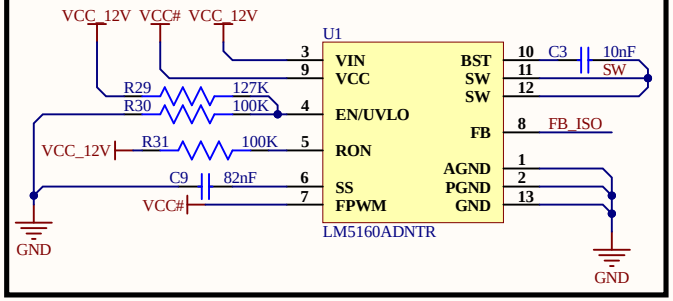

MOTOR SUPPLY
U_Motor-Supply-Section
Motor-Supply-Section.SchDoc
VS#
VCC_3V3_M
VCC_12V
VS
VS#
VCC_3V3_M
VCC_12V
VS
GND
GND


VS
VCC_5V
VDD_3V3
VS#
VCC_3V3_M
VCC_12V
GND
VS
VCC_5V
VDD_3V3
VS#
VCC_3V3_M
VCC_12V
GND

DECOUPLING CAPS

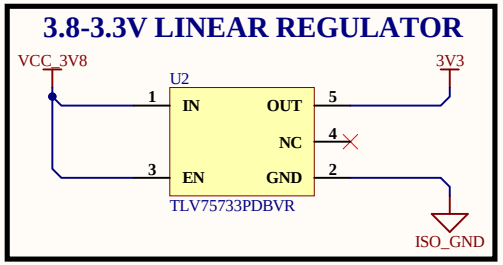


12V-3.8V ISOLATED BUCK CONVERTER

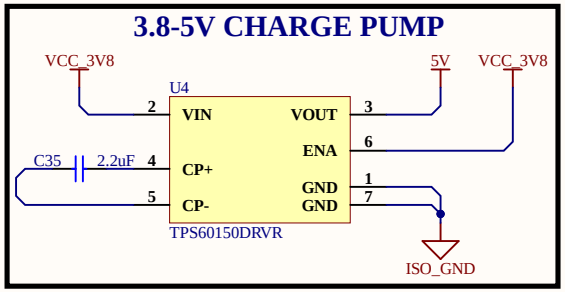


Primary side output voltage would be 5.7 for a 3.8V VSec (NP/Ns=Vpri/Vsec)
FB resistors RF2=3.7K and RF1=2K.
Max current of this transformer is 840mA. (suggested in datasheet)

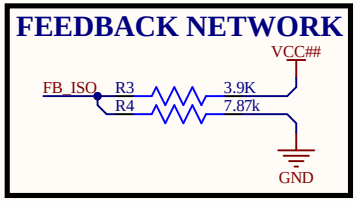
3.8-3.3V LINEAR REGULATOR



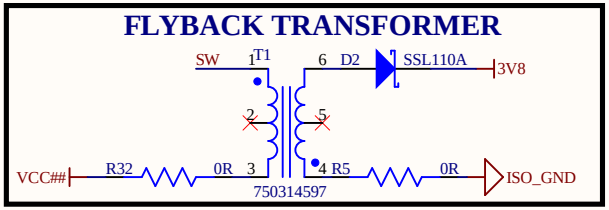
3.8-5V CHARGE PUMP



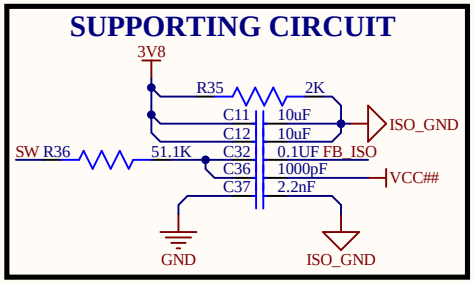
FEEDBACK NETWORK



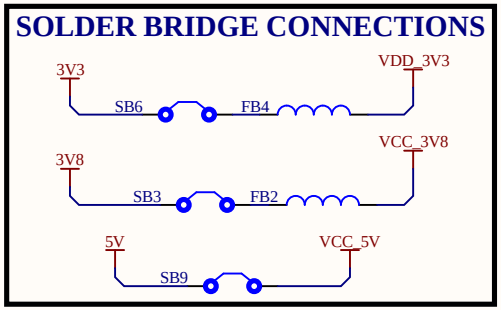
FLYBACK TRANSFORMER



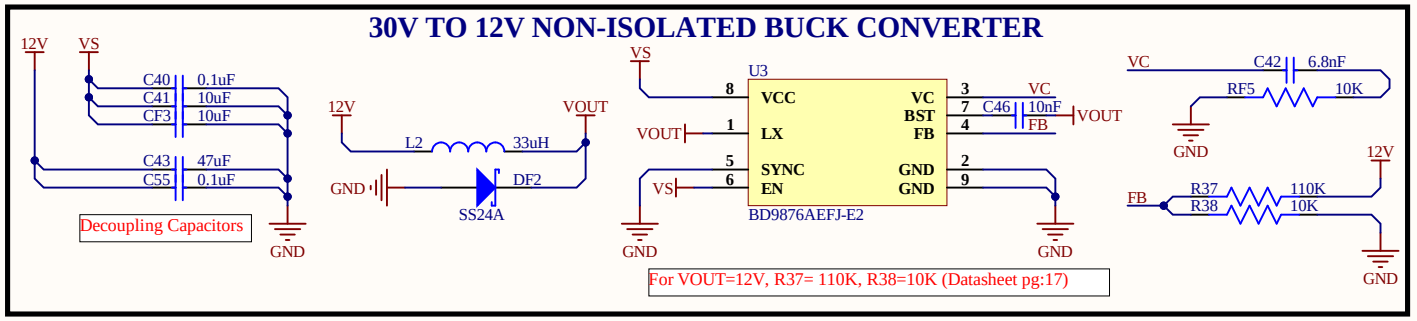
SUPPORTING CIRCUIT

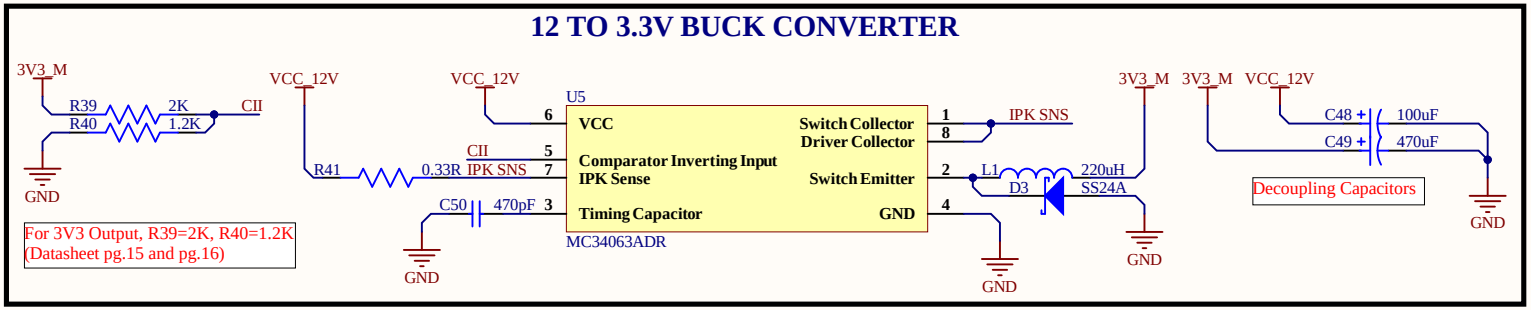


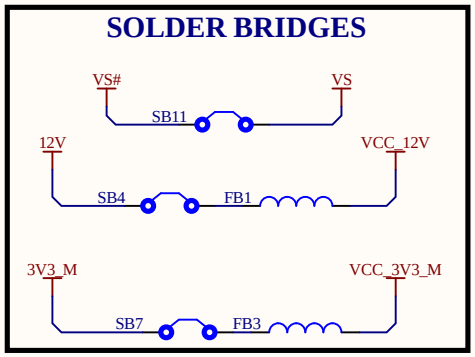
SOLDER BRIDGE CONNECTIONS

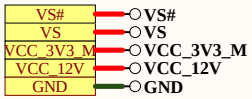


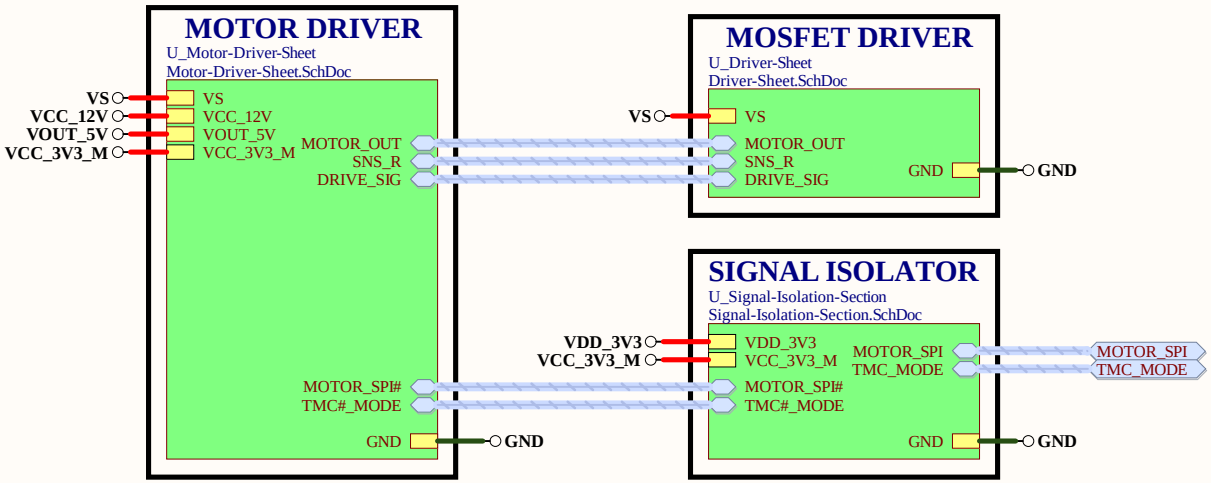
VCC_12V — VCC_12V
VCC_5V — VCC_5V
VDD_3V3 — VDD_3V3
GND — GND

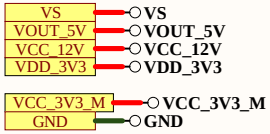

30V TO 12V NON-ISOLATED BUCK CONVERTER
12V
VS
C40 0.1uF
C41 10uF
CF3 10uF
C43 47uF
C55 0.1uF
Decoupling Capacitors
GND
12V
VOUT
L2 33uH
GND
DF2
SS24A
VS
U3
8 VCC
1 LX
5 SYNC
6 EN
VC 3
BST 7
FB 4
GND 2
GND 9
C46 10nF
VOUT
FB
BD9876AEFJ-E2
GND
GND
VC
C42 6.8nF
RF5 10K
GND
FB
R37 110K
R38 10K
12V
GND
For VOUT=12V, R37= 110K, R38=10K (Datasheet pg:17)

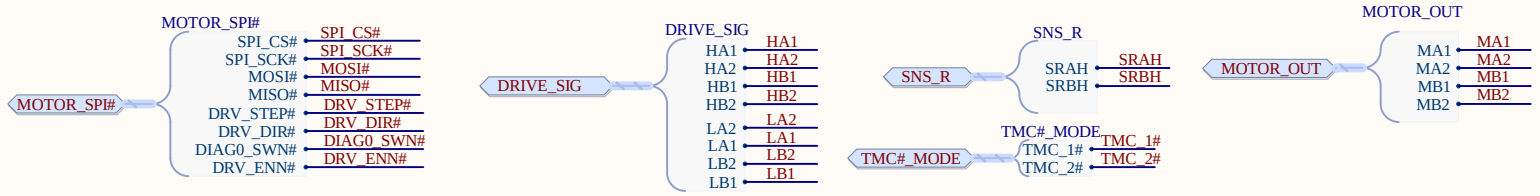

12 TO 3.3V BUCK CONVERTER
3V3_M
R39 2K
R40 1.2K
CII
GND
For 3V3 Output, R39=2K, R40=1.2K
(Datasheet pg.15 and pg.16)
VCC_12V
R41 0.33R
IPK SNS
CII
C50 470pF
GND
VCC_12V
U5
6 VCC
5 Comparator Inverting Input
7 IPK Sense
3 Timing Capacitor
Switch Collector 1
Driver Collector 8
Switch Emitter 2
GND 4
MC34063ADR
IPK SNS
L1 220uH
D3 SS24A
GND
GND
3V3_M
3V3_M
VCC_12V
C48 100uF
C49 470uF
Decoupling Capacitors
GND


SOLDER BRIDGES
VS#
VS
SB11
12V
VCC_12V
SB4
FB1
3V3_M
VCC_3V3_M
SB7
FB3

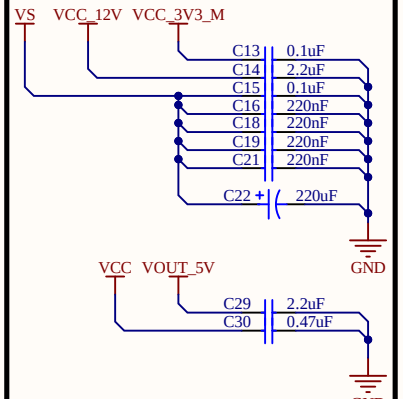

VS#
VS
VCC_3V3_M
VCC_12V
GND
VS#
VS
VCC_3V3_M
VCC_12V
GND


MOTOR DRIVER
U_Motor-Driver-Sheet
Motor-Driver-Sheet.SchDoc
VS
VCC_12V
VOUT_5V
VCC_3V3_M
MOTOR_OUT
SNS_R
DRIVE_SIG
MOTOR_SPI#
TMC#_MODE
GND
MOSFET DRIVER
U_Driver-Sheet
Driver-Sheet.SchDoc
SIGNAL ISOLATOR
U_Signal-Isolation-Section
Signal-Isolation-Section.SchDoc
VDD_3V3
MOTOR_SPI
TMC_MODE

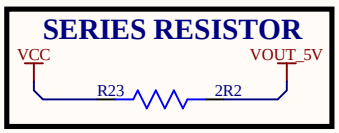

VS
VOUT_5V
VCC_12V
VDD_3V3
VCC_3V3_M
GND



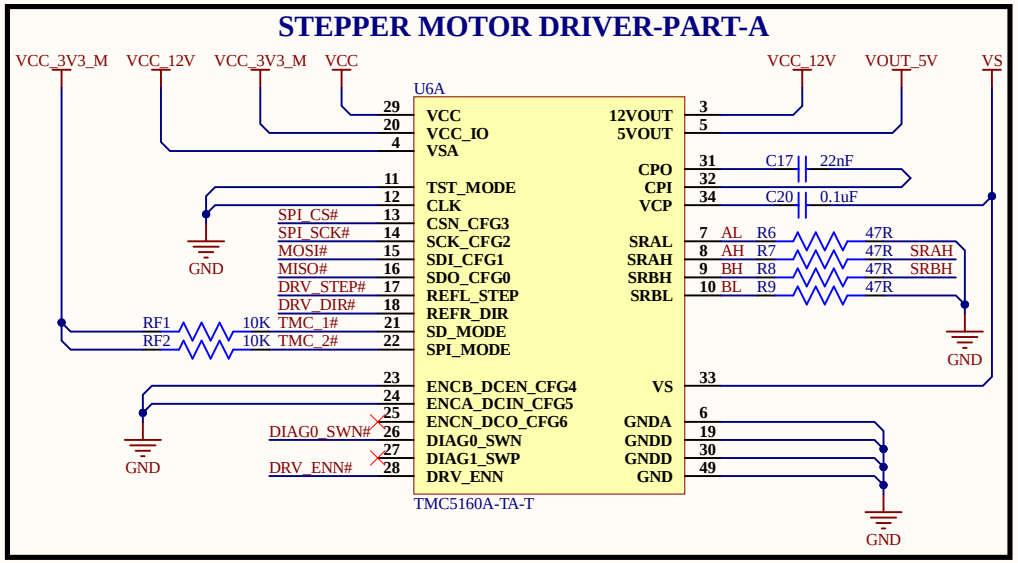
DECOUPLING CAPS



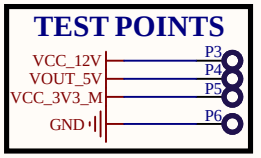
SERIES RESISTOR



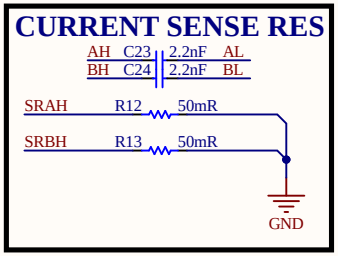
STEPPER MOTOR DRIVER-PART-A



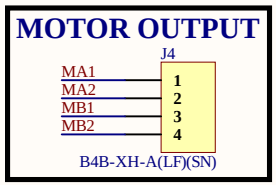
TEST POINTS



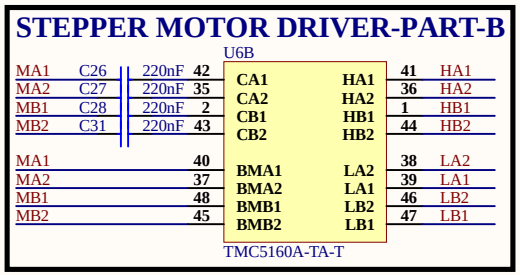
CURRENT SENSE RES



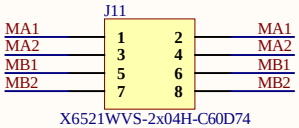
MOTOR OUTPUT

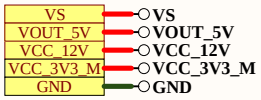


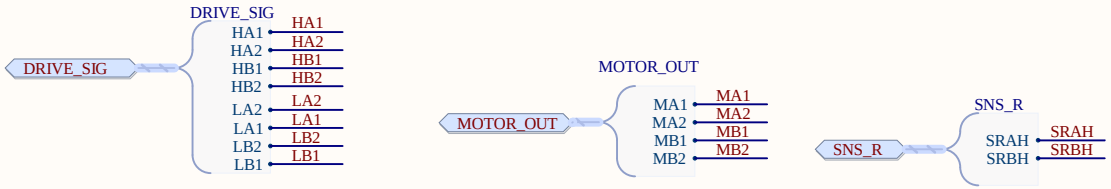
STEPPER MOTOR DRIVER-PART-B



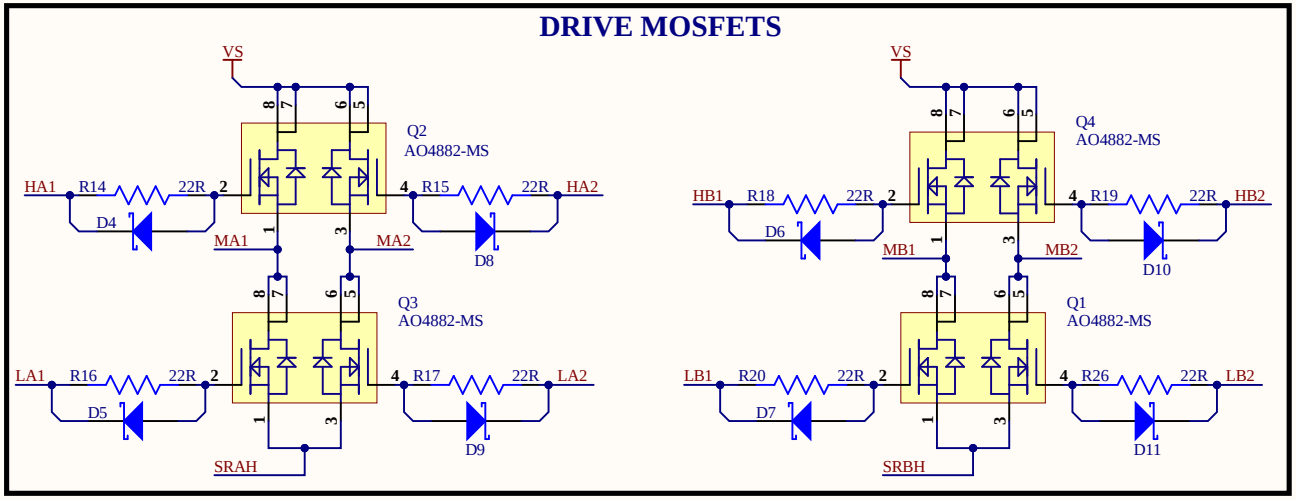
MOTOR OUTPUT FOR NEMA34



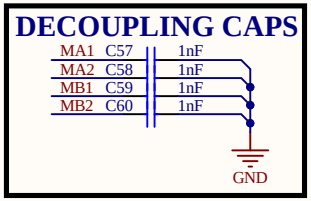




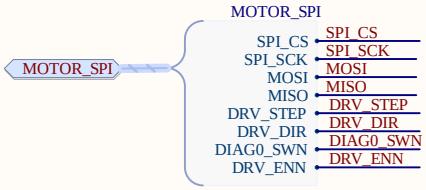
DRIVE MOSFETS

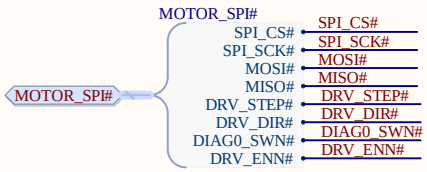


DECOUPLING CAPS



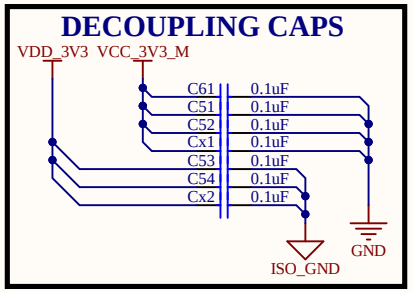


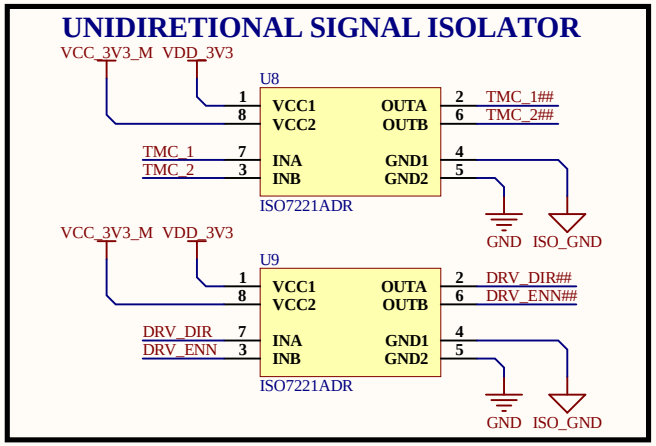


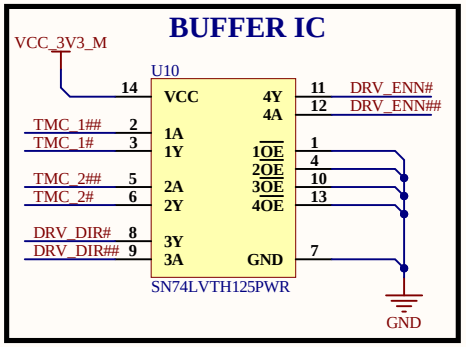


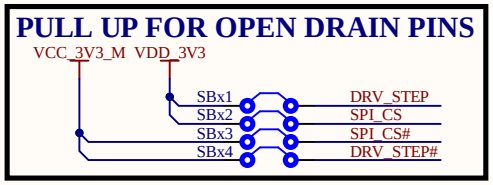




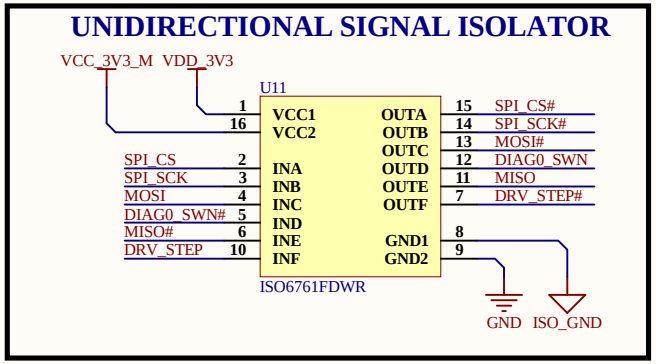








Mount 1K 0603 resistors on these bridges if needed

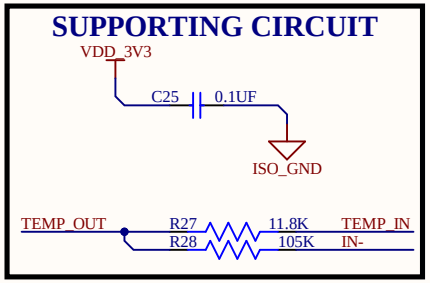




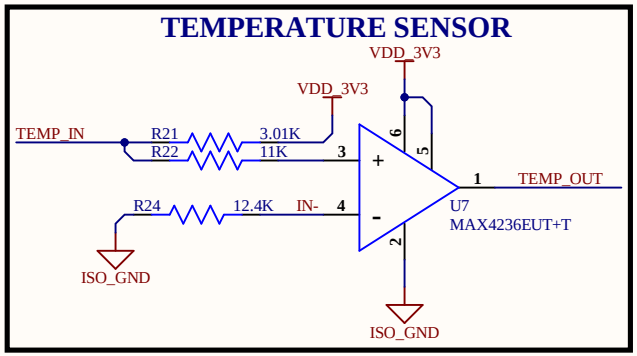
Title		
Size: A4	Number	Revision
Date: 5/09/2023		Sheet of
File: C:\Users\..\Signal-Isolation-Section.SchDoc		Drawn By:

TEMP_OUT TEMP_OUT

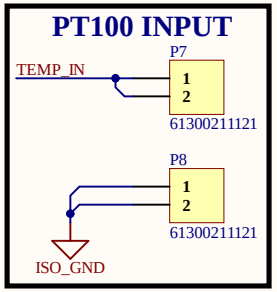
SUPPORTING CIRCUIT



TEMPERATURE SENSOR



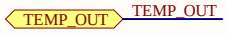
PT100 INPUT

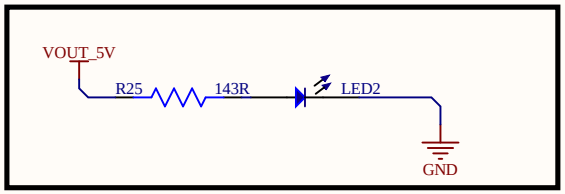


VDD_3V3 VDD_3V3
GND GND

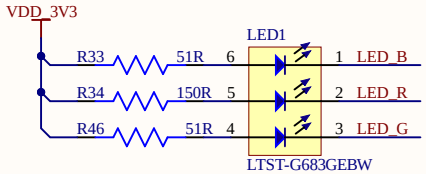
INDICATION FOR MOTOR 5V SUPPLY



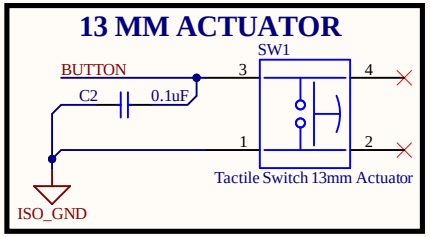




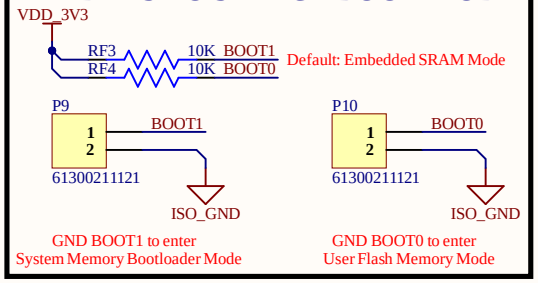
3.3V RGB LED

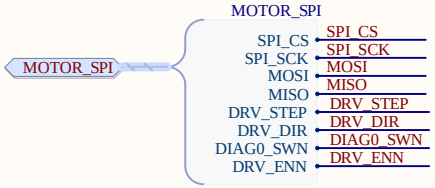


13 MM ACTUATOR

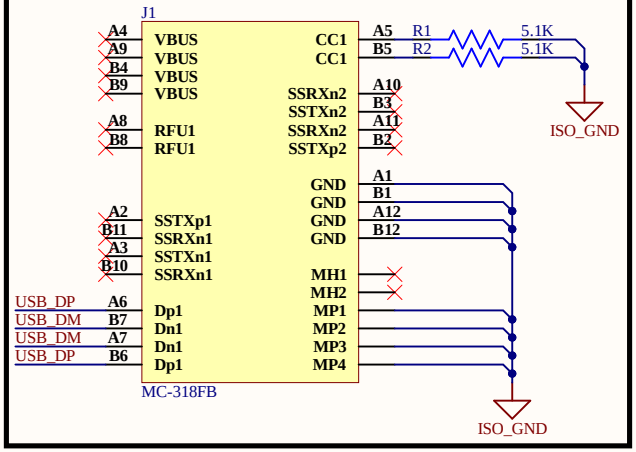


HEADERS TO SELECT BOOT MODE

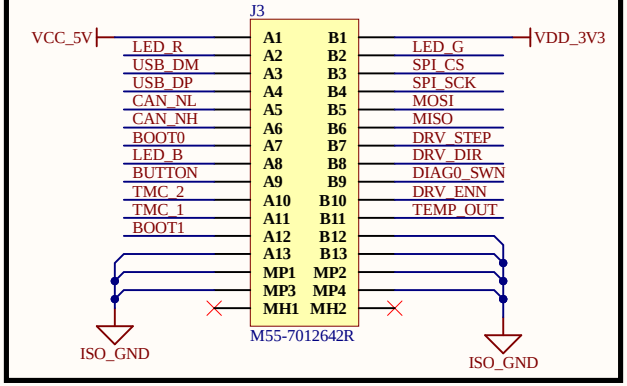




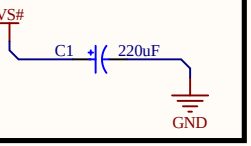
USB TYPE C FOR DATA TRANSFER



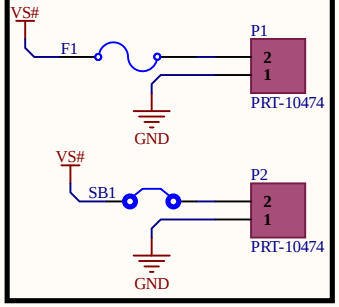
BOARD TO BOARD CONNECTOR



DECOUPLING CAP



INPUT CONNECTORS



RJ11 FOR CAN BUS

